Q1. Personalized Login Greeting

```
// Store the user's name and the current hour.
const userName = "Satyam";
const currentHour = new Date().getHours();

let greeting;

if (currentHour < 12) {
    greeting = `Good Morning ${userName}!`;
} else if (currentHour <= 17) {
    greeting = `Good Afternoon ${userName}!`;
} else {
    greeting = `Good Evening ${userName}!`;
}

console.log(greeting);
```

Q2. Multi-Type Data Summary

```
// Create values of different JavaScript data types.
const userName = "Satyam";
const age = 20;
const isStudent = true;
const subjects = ["JavaScript", "HTML", "CSS"];
const student = { name: "Satyam", course: "CSE" };
const middleName = null;
let address; // undefined because no value is assigned

// Array.isArray() identifies arrays correctly.
const report = [
    { Label: "userName", Value: userName, Type: typeof userName },
    { Label: "age", Value: age, Type: typeof age },
    { Label: "isStudent", Value: isStudent, Type: typeof isStudent },
    { Label: "subjects", Value: subjects.join(", "), Type: Array.isArray(subjects) ? "array" : typeof subjects },
    { Label: "student", Value: JSON.stringify(student), Type: typeof student },
    { Label: "middleName", Value: String(middleName), Type: middleName === null ? "null" : typeof middleName },
    { Label: "address", Value: String(address), Type: typeof address }
];

console.table(report);
```

Q3. Monthly Expense Tracker

```
// Store monthly expenses for five categories.
const expenses = {
    food: 4500,
    travel: 1200,
    rent: 8000,
    bills: 1800,
    leisure: 1500
};

let total = 0;

// Add each category amount to the total.
for (const category in expenses) {
    total += expenses[category];
}

const average = total / Object.keys(expenses).length;
const taxRate = 0.10;
const tax = total * taxRate;
total += tax; // Add 10% tax using an assignment operator.

console.log(`Total before tax: Rs. ${(total - tax).toFixed(2)}`);
console.log(`Average expense: Rs. ${average.toFixed(2)}`);
console.log(`Tax (10%): Rs. ${tax.toFixed(2)}`);
console.log(`Final amount after tax: Rs. ${total.toFixed(2)}`);
```

Q4. Academic Performance Evaluator

```
// Marks are out of 100 for five subjects.
const marks = [88, 76, 91, 84, 79];

const totalMarks = marks.reduce((sum, mark) => sum + mark, 0);
const average = totalMarks / marks.length;
const percentage = (totalMarks / (marks.length * 100)) * 100;
const hasFailedSubject = marks.some(mark => mark < 35);

let result;

if (hasFailedSubject) {
    result = "Detained";
} else if (percentage >= 85) {
    result = "Promoted with Distinction";
} else if (percentage >= 50 && percentage < 85) {
    result = "Promoted";
} else {
    result = "Detained";
}

console.log(`Average marks: ${average.toFixed(2)}`);
console.log(`Overall percentage: ${percentage.toFixed(2)}%`);
console.log(`Result: ${result}`);
```

Q5. Weather Activity Planner

```
// Sample weather values; change them to test other conditions.
const temperature = 28;
const isRaining = false;
const windSpeed = 12;

let advice;

if (isRaining) {
    advice = "Stay indoors with hot coffee.";
} else if (temperature > 35) {
    advice = "Go swimming.";
} else if (temperature < 15 && windSpeed > 20) {
    advice = "Too cold and windy — stay home.";
} else if (temperature >= 15 && temperature <= 35 && !isRaining && windSpeed <= 20) {
    advice = "Perfect day for a walk.";
} else {
    advice = "Check the weather and plan carefully.";
}

console.log(advice);
```

Q6. Progressive Discount System

```
// Enter the purchase amount here.
const total = 7500;

let discountRate;

if (total >= 10000) {
    discountRate = 0.25;
} else if (total >= 5000) {
    discountRate = 0.15;
} else if (total >= 2000) {
    discountRate = 0.05;
} else {
    discountRate = 0;
}

const discountAmount = Math.round(total * discountRate);
const finalPrice = Math.round(total - discountAmount);

console.log(`Original total: Rs. ${total}`);
console.log(`Discount: ${discountRate * 100}%`);
console.log(`Discount amount: Rs. ${discountAmount}`);
console.log(`Final price: Rs. ${finalPrice}`);
```

Q7. Smart Guessing Game (Number Range)

```
// Generate a secret integer from 1 to 50.
const secretNumber = Math.floor(Math.random() * 50) + 1;
const userGuess = 27; // Change this value to test different guesses.

if (userGuess === secretNumber) {
    console.log("Correct guess!");
} else {
    const difference = Math.abs(userGuess - secretNumber);

    if (difference <= 3) {
        console.log("Very close!");
    } else if (userGuess > secretNumber) {
        console.log("Too high");
    } else {
        console.log("Too low");
    }
}

console.log(`Secret number: ${secretNumber}`);
```

Q8. Employee Salary Projection

```
// Set the current annual salary and increment rate.
let salary = 500000;
const annualIncrementRate = 8; // percentage
const projection = [];

for (let year = 1; year <= 5; year++) {
    salary += salary * (annualIncrementRate / 100);
    salary = Math.round(salary);

    projection.push({
        Year: year,
        "Projected Salary (Rs.)": salary
    });
}

console.table(projection);
```

Q9. Random Math Quiz Generator

```
// Generate two random integers between 1 and 20.
const number1 = Math.floor(Math.random() * 20) + 1;
const number2 = Math.floor(Math.random() * 20) + 1;
const operators = ["+", "-", "*", "/"];
const operator = operators[Math.floor(Math.random() * operators.length)];

let answer;

switch (operator) {
    case "+":
        answer = number1 + number2;
        break;
    case "-":
        answer = number1 - number2;
        break;
    case "*":
        answer = number1 * number2;
        break;
    case "/":
        answer = number1 / number2;
        break;
}

if (operator === "/") {
    answer = answer.toFixed(2);
}

console.log(`Question: ${number1} ${operator} ${number2} = ?`);
console.log(`Correct answer: ${answer}`);
```

Q10. Citizen Eligibility Validator

Note: The given conditions overlap for citizens aged 18–20. This solution treats ages 21+ as eligible for all services and ages 18–20 as eligible to vote only.

```
// Example input values; change them to test other cases.
const age = 19;
const isCitizen = true;

if (isCitizen) {
    if (age >= 21) {
        console.log("Eligible for all services.");
    } else if (age >= 18 && age <= 20) {
        console.log("Eligible to vote only.");
    } else {
        console.log("Not eligible yet.");
    }
} else {
    if (age >= 18) {
        console.log("Only age criteria met.");
    } else {
        console.log("Not eligible yet.");
    }
}
```